



Course: BCE 412 MACHINE LEARNING

Practical: Data Preprocessing, Feature Engineering, NLP, and Reinforcement Learning Techniques

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Objective

The purpose of this practical is to enable students to apply key data science techniques including dimensionality reduction, data preprocessing, feature engineering, natural language processing, and reinforcement learning using real-world datasets.

Task

You are required to:

1. Download a dataset of your choice from the following platforms:

Dataset Sources

- Hugging Face: <https://huggingface.co/datasets>
- Kaggle : <https://www.kaggle.com/datasets>
- UCI: <https://archive.ics.uci.edu/ml/index.php>

Suggested Datasets (You may choose one)

- IMDB Movie Reviews (NLP)
<https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews>
- Titanic Dataset (Structured Data)
<https://www.kaggle.com/datasets/yasserh/titanic-dataset>
- Wine Quality Dataset (PCA & Feature Engineering)
<https://archive.ics.uci.edu/ml/datasets/wine+quality>
- Amazon Reviews (Text Mining)
https://huggingface.co/datasets/amazon_reviews_multi
- CartPole Environment (Reinforcement Learning)
https://www.gymnasium.dev/environments/classic_control/cart_pole/

Required Tasks

1. Data Cleaning

- Handle missing values (imputation or removal)
- Remove duplicates
- Handle outliers
- Convert data types where necessary

2. Feature Engineering

- Create new features
- Encode categorical variables
- Normalize or standardize data

3. Principal Component Analysis (PCA)

- Apply PCA for dimensionality reduction
- Explain variance retained
- Visualize results

4. Bag of Words (NLP Task)

(If dataset contains text)

- Clean text data
- Tokenize text
- Build Bag of Words model
- Display feature matrix

5. Reinforcement Learning

- Implement a simple model (e.g., Q-learning)
- Define:
 - States
 - Actions
 - Rewards
- Demonstrate learning process

Implementation Requirements

- Use Python
- Use Google Colab for implementation
- Write clean, well-commented code

Submission Requirements

1. Google Colab Notebook

- Provide a shareable link
- Ensure access is enabled (*Anyone with the link can view*)

2. Summary Report (3–5 pages)

Include:

- Introduction (Dataset description)
- Methodology (All techniques used)
- Results (Outputs, charts, insights)
- Discussion (Interpretation, challenges)
- Conclusion

Evaluation Criteria

- Data preparation – 20%
- Implementation – 40%
- Code quality – 15%
- Report quality – 25%

Important Notes

- Dataset must be well prepared before analysis
- Ensure all five techniques are implemented
- Provide clear explanations and visualizations
- Include Colab link in submission